IoT Based Home Automation System Using ThingSpeak

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***Abstract*—This paper presents the motivation behind the idea of home automation based on Internet of Things (IoT). The aim of this research is to develop an IoT based system that provides remote access to the user from an android mobile application to control electrical appliances. Furthermore, it is focusing on proposing a solution which will help cut down the energy consumption by following automated behavior which it is programmed to do. Use of energy consumption can be monitored through android mobile application. System is easy to operate and above all, it is user friendly. For the implementation and testing of this model, the targeted appliances are interior and exterior lights, fans, water sprinkler for lawn and door locks. Main hardware components used are Arduino UNO controller and Raspberry Pi Model 3 for development which uses the services provided by ThingSpeak Cloud. The results are analyzed based on the reports generated from the calculated amount of energy consumption when devices were not connected with the system as compared to when they are connected.**

Keywords— Internet of Things (IoT), Raspberry Pi Model 3, Arduino UNO, ThingSpeak.

I. INTRODUCTION

Internet of Things (IoT) is advanced addition of internet connectivity. It connects physical devices used in daily life with each other through the internet. This technology has become a bridge between multiple fields such as digital electronics, sensors, actuators and internet connectivity. Devices which are connected with each other via the internet can communicate and interact. Furthermore, these devices provide easy controlling and monitoring. Within the recent years, [1] rapid development in Internet of Things (IoT) technology has fascinated the world to use it in a number of fields (e.g. Farming, Education, Industry and Home Automation etc.) [2]. Home automation has become a popular idea in development of human comfort, particularly for the private houses because of the need for comfortable living and reliability. There are many gadgets in the market for home automation purpose. Moreover, controlling a house with wireless technology can provide high level of feasibility. These systems can help to control and cut down the electricity consumption by controlling appliances automatically. One can keep track of their day to day power consumption [3]. This system is not just designed for electric appliance control, IoT can be used for security [4] and other standalone modules such as automated garden water sprinkling and exterior lights. These modules can be monitored through an Android mobile/tablet device. A lot of research has been done to develop such systems. Data is retrieved from sensors which generate massive information [5] and is processed to achieve and present future predictions.

This research is formulated to increase the knowledge of development and implementation of the smart home automation system using cloud service. It is an automatic system which can work 24/7. An Android mobile application is developed which is used for the remote controlling of home appliances and locking/unlocking of door locks from anywhere after establishing a successful connection with Internet through Wi-Fi or 3G/4G. Moreover, one can check their periodic power consumption through the application. This system also has a standalone module which deals with the automation of lawn water supplies.

This paper is organized as follows. The next section consists of literature review. The architecture of proposed system is presented in Section III where functionalities of each module is presented along with the explanation of the system architecture. Section IV describes methodology in which all the configurations of cloud services, connections between the hardware modules, dataflow among those modules and internal implementation concept of android mobile application are presented in detail. Experimental results have been represented in Section V with the help of a graph for a quick interpretation and understanding of the results. Finally, this paper is concluded in Section VI along with the importance underlying the future scope of this research.

II. LITERATURE REVIEW

There are a number of home automation solutions available in the market as well as proposed in literature. One approach is providing the controls of home appliances, efficient power consumption and house monitoring using the google Cloud Services and ZigBee. Providing remote interactions and monitoring through a Web Application Client and Mobile application. Visualization of the atmosphere of the environment (e.g. Temp and Humidity) [6].

An email based switching system is also proposed to control the electric appliances. Using a Raspberry pi, an algorithm in python that reads the command written in the title of the email can send the feedback after executing those commands [7]. Email based switching is less efficient as compared to the direct signal transmission. Email

transmissions may have lesser reliability such as syntax error in the command which can lead to an error or the client can assume such emails as spam.

Mandula et al. [8] has proposed two prototypes, one for indoor and another for the outdoor using Arduino. Indoor appliances are controlled with Bluetooth and outdoor with Ethernet over IP Addresses. The system is able to control home appliances like bulbs, fans, air conditioner and LED TV etc. Appliance's ON and OFF switching is provided using the Android mobile application. As the data is not being saved in any database, it is impossible to analyze or keep track of energy consumption and any other automatic intelligent response for automation.

Using ZigBee, WLAN and IEEE 802.11 Technology [9] for the development of such system could be another way in approaching it, as discussed by Khan et al. [10]. Common household appliances like Washing Machine, Air Conditioner etc. can be controlled easily and measurement of their power consumption by monitoring the usage cycle. System [10] was tested using computer simulator. Also, proposed scheduling mechanism to minimize the energy consumption for solar grids. A CoZNET [11] mechanism system which does not effect by the interferences of Wi-Fi and wireless sensor network [10]. System was proposed to demonstrate the application of smart grid of which the power consumption could be observed. It has automated scheduling of switching the household appliances and most importantly a system which does not get effected by the interferences of other electromagnetic waves as they can affect the performance and efficiency of the system.

One of the method by D'souza et al. is implemented for basic home automation system using Raspberry pi. The controlling of the electric appliances is through mobile application and the visualization of the system processes is through Web application page. Automatic switching is done in the result of sensor feedback [12].

Alternative prototype is developed using Edge Computing (EC) [13] and IoT based system is controlled using android mobile application. EC is developed in NodeJS using Raspberry pi. Connected appliances are lights, door locks, air conditioners and security cameras [14]. Only controls provided are the switching ON/OFF for the appliances on the detection of sensor's feedbacks.

Now a days, the place of simple automation systems have been taken by the intelligent systems. A proposed system by Zainab et al. [15] can be observed which is designed for smart cities. This system is designed with the help of IoT and Artificial Intelligence (AI). Presented system claims that, it will help reducing the congestions of traffic on roads.

Just like an intelligent system for traffic, Safeeullah et al. [16] have highlighted the IoT based healthcare. It is a vast project which includes many small systems such as patient fall detection, medical fridges, patient surveillance and many more which can be monitored, controlled and maintained. This model has analyzed all the data using Fuzzy Analytical Hierarchy Process (FAHP) [16].

For this research paper, we have specially focused on proposing the system for Home Automation. The system can be expanded as required for the future research point of view.

III. System Architecture

In this section, we have provided the architecture of our suggested IoT Based Home Automation System. The architecture consisted of multiple modules named as Android mobile application, ThingSpeak Cloud services, Raspberry pi Model 3, Arduino Uno and sensors. Fig. 1 below presents the proposed system architecture.

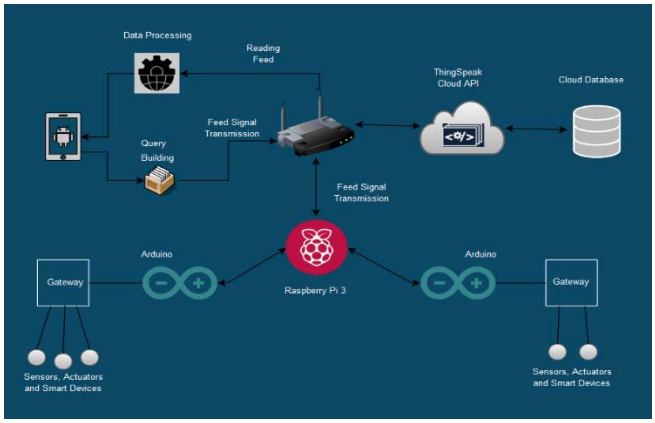


Fig. 1. System Architecture

It provides the clear concept about the implementation of the system. The descriptions of the used devices are given below:

1) Android Mobile Application

Android mobile application provides the interface for the user to interact with the system and controls the appliances. User can switch ON and OFF the electric appliances. One can LOCK and UNLOCK the doors. Moreover, the user can view the representation of data on the mobile screen. Represented data is the consumption of the electric energy consumed by the individual appliance. Data is retrieved from the ThingSpeak cloud database and processed inside the application and showed on the screen.

2) ThingSpeak

ThingSpeak [17] is an open source cloud and analytics platform which provides the services to analyze and visualize the data in real time. This cloud interacts with both the mobile application and the Raspberry pi. We are using this cloud service for signal communication and signal data storage. Stored data is fetched from database of the ThingSpeak, processed and results generated are shown on Android mobile application.

3) Raspberry pi

In this system, Raspberry pi Model 3 is used as a bridge between ThingSpeak and Arduino. Raspberry pi gets the signal from the cloud and based on that signal the instruction is sent to the specific Arduino to trigger the related appliance.

4) Arduino

For the development of this system Arduino UNO is used. Arduino plays its role as an actuator for Raspberry pi. Arduino receives the signals from Raspberry pi and triggers the appliances. Arduino also works as a standalone module which manages the responses between sensors and actuators. Feedback signals and sensor data is sent back to the Raspberry pi from Arduino to ensure the acknowledgement of successful signal transmission. The idea behind using a Raspberry pi is to provide the client server model. By using this configuration

we can attach more than one Arduino with a single Raspberry pi to extend the system.

5) Circuitry

Following circuits in Fig. 2 and Fig. 3 represent the circuit diagrams of modules. These circuits are the conceptual prototypes and real components could be different in appearance. Fig. 2 contains the circuit of exterior lights and water sprinkler. Soil moisture sensor detects the moisture in the soil and turns the water sprinkler ON and OFF according to the value of threshold. Moreover, the LDR sensor detects the intensity of sun light and turns the exterior light ON/OFF depending on the sun light detection.

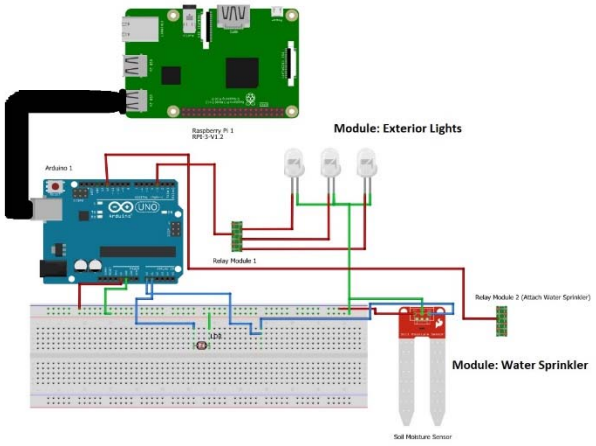


Fig. 2. Circuit Modules: Exterior Lights and Water Sprinkler

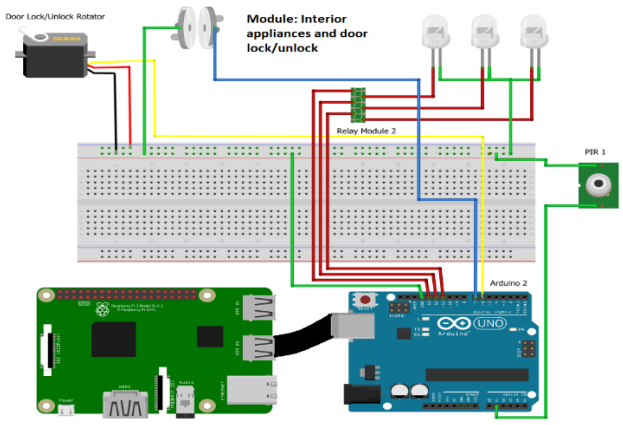


Fig. 3. Circuit Modules: Interior Lights and Door Lock/Unlock

In Fig. 3, the circuit of interior lights and door lock/unlock has been shown. In this circuit, PIR sensor detects the motion inside the room if anyone is present in the room and if not, then the room system automatically turns off the interior appliances of the room to save energy. Magnetic read switch verifies if the door is properly closed or not. If the door is closed the servo motor will move the spindle to lock the door. On the contrary, if the door is not closed, the read switch will generate an error to let the system know that the door cannot be locked as it is not closed properly.

IV. Methodology

An applied methodology has been proposed for this system. A system is developed to simulate the features provided as follows:

A. Features of the proposed System

- Maintains the switching signal log over the cloud.
- Use of signals to trigger the connected appliance.
- Provide remote control access to the user from an Android mobile device.
- Show the usage of energy consumption according to the time period selected.
- Provide automated exterior light switching and maintains the log.
- Control the lawn water supplies through automated sprinkler.

B. Used Components

There are many ways and components in the market that can be used alternatively for the development of such systems. Table I consists of the components we have used to develop our proposed system due to the ease of their availability and massive range of functionality, features and documentation. Similar components can also be used as the replacement for components mentioned below. The quantity of components shown in table is as per suggested model.

TABLE I. USED COMPONENTS

Sr.	Component	Quantity	Use
1	Android Device	1	User Interface
2	Raspberry pi model 3	1	Main IoT controller
3	Arduino	2	Controllers for Raspberry Pi as actuators
4	Magnetic Read Switch	1	Door open/close detection
5	PIR Sensor	1	Motion Detection inside the room
6	LDR Sensor	1	Light Detection for exterior lights
7	Moisture Sensor	As required	Moisture detection in the soil
8	Relay Module	As required	Gateway from Arduino to electric appliances
9	Servo Motor	1	Lock spindol rotator

C. System Implementation

In this section, we have discussed the implementation of the system. Implementation starts with the configuration of ThingSpeak Cloud.

1) Cloud Configurations

The cloud used for this system is ThingSpeak. In order to use its services, a signup is required after which we need to register an application on ThingSpeak. After registration of application, we make channels for the application. Each channel represents a system. Within the channels, there are fields which we can name according to our requirement. In a channel, each field represents the feed signal transmission slot for their corresponding attached appliance.

Fig. 4, 5 and 6 are the representation of the feed data in the channel for bulb1, fan and exterior lights respectively. In cloud database, signals are represented as 1 = ON and 0 = OFF. These signals also serve as logs with time stamps.

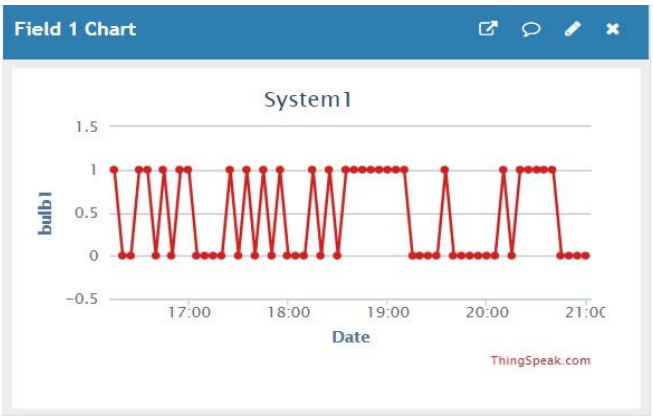


Fig. 4. ThingSpeak Feed Data Signals for Bulb1

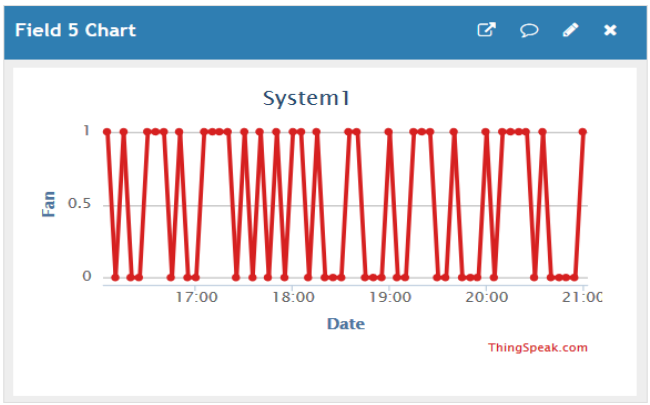


Fig. 5. ThingSpeak Feed Data Signals for Fan

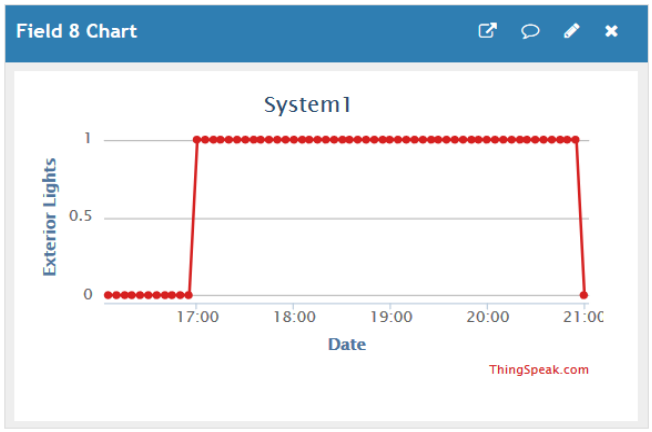


Fig. 6. ThingSpeak Feed Data Signals for Exterior Lights

We can access the services using the ThingSpeak API. The API service uses GET requests for the execution of the query. These queries are generated from Android Mobile devices as well as Raspberry pi to send signals and maintain the log in the cloud database.

There are four main components for query building and feed data download/upload. The components are as follows:

Link = "http://api.thingspeak.com/"

Field = "#" (number)

Channel = "######" (number)

API_KEY="###AA##AAAA##AAA"

Time_Zone = "Asia/Karachi" (Optional)

2) Raspberry pi and Arduino Configurations

It is very important to configure Raspberry pi and Arduino correctly. There could be many ways to connect these components with each other (i.e. Wi-Fi, Bluetooth, GPIO and Serial Communication). We have used serial communication in this system.

Raspberry pi continually accesses the signals stored on the cloud database and based on the signals it passes the instructions to the Arduino to trigger the specific actuator according to the signal received. Besides this, Arduino also works as a standalone module whenever data from a sensor is captured, it triggers the response accordingly and sends the signals back to the Raspberry pi as an acknowledgement and to upload the log on cloud. A lawn water sprinkler is also attached with the Arduino to manage the water supply for the lawn as a standalone.

3) User Interaction

Android Mobile application is the main component for the interaction between a user and the system. Application provides the remote controls in order to control electric appliances and door locks from all over the world through the internet.

There are two main features in the application. Firstly, it provides the remote access to the user and secondly, it shows the use of energy consumption. In order to check the energy consumption, the user needs to select the connected electric appliance and time duration of which the user wants to know the consumption for.

The query is executed after the accomplishment of a secure internet connection which could be through Wi-Fi or Mobile Data (3G/4G). When the application has to present the energy consumption, it downloads the CSV file from the cloud and processes it. After processing and extracting the information, the energy consumption is showed in kWh on the screen. Kilowatt hour [18] is the unit of Energy and Kilowatt is the unit of power. Power can be calculated with the following formula.

$$Power = \frac{Energy}{time} \qquad (1)$$

Energy can be measured by rearranging the equation "(1)". If we have the power in kilowatt and time in hours then we can calculate the energy in kilowatt hour using "(2)". Energy consumption is represented in kWh in the application.

$$Energy(KWh) = Power(KW)\ x\ time(Hour) \qquad (2)$$

4) System Data Flow

Fig. 7 represents the data flow of the system. It also represents user interaction scenario where the user just has to use mobile application in order to interact with the system. Mobile application directly connects to the cloud and inserts data along with retrieving the feed data. Rest of the system is connected with the cloud and required data download and upload are through the Raspberry pi. The Arduino Client gets interactions from the Raspberry pi and manages the appliances and their acknowledgements. Arduino works as a client interacting with the Raspberry pi and also as a standalone module which responds accordingly to the sensor feedback data

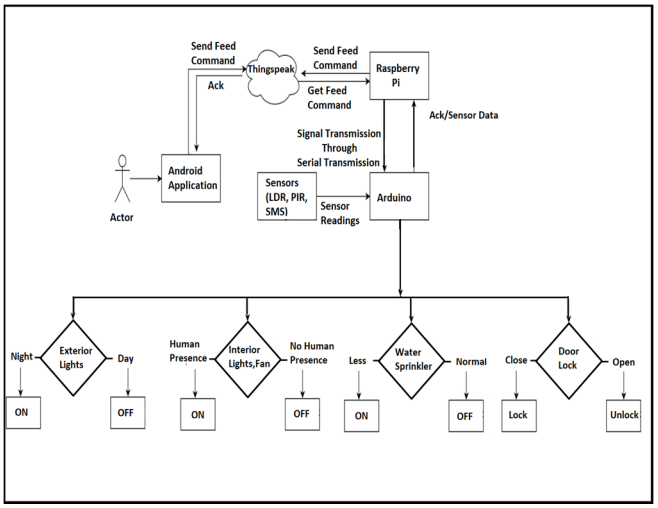


Fig. 7. System Data Flow Diagram

V. Results

This section presents the results which were gathered after the successful implementation of the system. We implemented this system in start of the January 2019 till the end of the May 2019. We tested our mobile application and observed the response in the switching ON/OFF the appliances and the door Lock/Unlock. Data log and signal transmission is monitored through the web portal provided by ThingSpeak cloud also showed in figure "Fig. 4", "Fig. 5" and "Fig. 6". Two observations were noted down to compare our projected system's results. In the following graph, decrease in the energy consumption can be observed. The graph is representing the comparisons between two scenarios one is with the system implemented and the other is without the system implementation. Bars in "Blue" color represents the readings of energy consumption with proposed system and the bars in "Orange" are the representation of simple readings without system implementation.

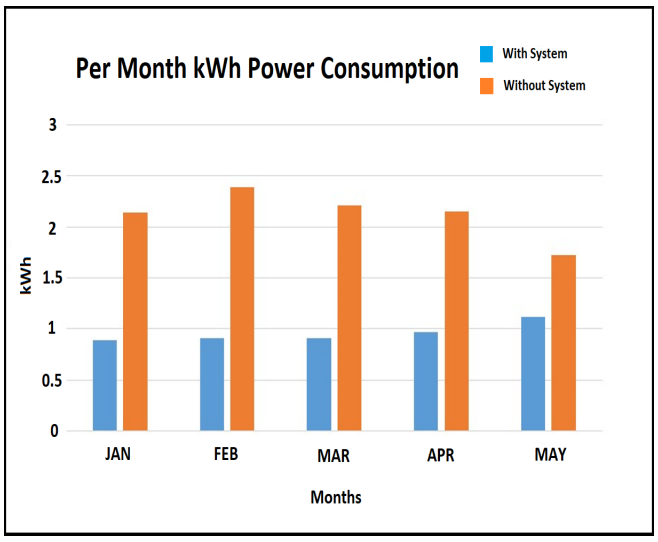


Fig. 8. Results

The graph in Fig. 8 shows our results from which we can conclude that the system has worked as expected to decrease the energy consumption. It can be noticed that the system is responsible and was able to decrease the energy consumption as compared to the simple energy consumption without system.

VI. Limitations

Anticipated system has some limitations regarding its working. First of all, system needs continuous supply of electricity to stay in synchronization. System will not operate in the absence of electric supply. This limitation can be resolved by attaching a battery power supply which can provide the electric power backup in the absence of main electric power.

Secondly signal transmission and cloud interaction is not possible without the connectivity of Internet. This can also be resolved if we make sure that the system is connected with a valid, secure and uninterrupted internet connection provided by the internet service provider.

VII. Conclusion And Future Work

The proposed system is developed with the emphasis to decrease the energy consumption using Internet of Things (IoT) technology. System has shown realistic results in managing and minimizing the energy consumption. Suggested architecture is developed in keeping its budget as low as possible. This system can help to decrease massive amount of energy consumption if developed and implemented at a large scale for homes as well as for industries. Research and implementation of fuzzy algorithm for the management of electric energy can also be projected as an extension of this research. Use of fuzzy algorithm might affect the efficiency of the system. Another way to broaden the scope of this research could be the management of traffic lights and street lights as street lights consume massive amount of electric energy. If this concept is used at large scale for street lights of the city as well as a country, it will save a huge amount of energy consumption.

This research will be a milestone for future intelligent systems which will be able to adapt and learn with the time and use more advance machine learning and deep learning knowledge.

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